

Chapter 5 — Question 4

Group 4

Question

Suppose that we use some statistical learning method to make a prediction for the response Y for a particular value of the predictor X . Carefully describe how we might estimate the standard deviation of our prediction.

Answer

We have a single fitted model that produces one predicted value $\hat{Y}$ at a given value of X . The question is how confident we should be in that single prediction — that is, how much would $\hat{Y}$ likely change if we had trained our model on a different sample of data drawn from the same underlying population?

Since most statistical learning methods do not come with a closed-form formula for the standard error of a prediction (this is really only convenient for simple models such as ordinary least squares), the natural general-purpose tool is the **bootstrap**. The procedure is as follows.

Step 1. Resample with replacement. From the original training data set of size n , draw a new data set of size n by sampling observations *with replacement*. This is called a bootstrap sample. Because sampling is with replacement, some original observations may appear multiple times in the bootstrap sample, and others may not appear at all.

Step 2. Refit the model. Fit the statistical learning method on this bootstrap sample, exactly as it was fit on the original data.

Step 3. Predict at the value of interest. Using this newly fitted model, compute the predicted response at the specific value of X we care about. Call this prediction $\hat{Y}^{*1}$.

Step 4. Repeat many times. Repeat Steps 1–3 a large number of times, B (e.g. $B = 1,000$ or $B = 10,000$), each time drawing a fresh bootstrap sample, refitting the model, and predicting at the same X . This produces a collection of B bootstrap predictions, all at the same point X but coming from models fit on slightly different resampled training sets:

$$\hat{Y}^{*1}, \hat{Y}^{*2}, \dots, \hat{Y}^{*B}.$$

Step 5. Compute the spread of these predictions. The standard deviation of the B bootstrap predictions is our estimate of the standard deviation (standard error) of the original prediction $\hat{Y}$:

$$\widehat{\text{SD}}(\hat{Y}) = \sqrt{\frac{1}{B-1} \sum_{r=1}^B \left(\hat{Y}^{*r} - \overline{\hat{Y}^*} \right)^2}, \quad \text{where } \overline{\hat{Y}^*} = \frac{1}{B} \sum_{r=1}^B \hat{Y}^{*r}.$$

Interpretation

This quantity, $\widehat{\text{SD}}(\hat{Y})$, tells us how much the predicted value at X would typically vary if we had trained our model on a different random sample from the same population. A small value indicates the prediction is stable and trustworthy; a large value indicates the prediction is highly sensitive to the particular sample of training data we happened to observe, and should be treated with more caution — for example, by reporting a wider interval around the point prediction.

Why the Bootstrap is the Right Tool Here

The key advantage of this approach is that it requires *no assumptions* about the underlying distribution of the data or the mathematical form of the fitted model. It works identically whether the model is linear regression, a decision tree, a neural network, or any other learning method — as long as the model can be refit on a resampled data set and used to generate a prediction. This generality is precisely why the bootstrap, introduced in Section 5.2, is described as such a powerful and widely applicable tool: it converts the abstract question “how uncertain is this prediction?” into a concrete, computable quantity, simply by simulating the process of re-sampling from the population using the data we already have.